

Hybrid Mamba-Transformer Architectures for EEG-to-Text Decoding: From Character Classification to LLM-Corrected Sentence Production

A. L. Tito

M.Sc. Project, Department of Computer Science
(placeholder affiliation — replace before submission)
email: atito@example.edu

Abstract—Decoding written language production from non-invasive brain recordings is a central goal of brain-computer interface (BCI) research. The recent Brain2Qwerty system showed that a convolutional front-end coupled with a sentence-level Transformer and a character-level language model can decode typed sentences from magneto- and electroencephalography (M/EEG), but self-attention scales quadratically with sequence length and decoded characters remain error-prone without strong linguistic correction. In parallel, state-space models (SSMs) such as Mamba and Mamba-2 provide linear-time sequence modeling, and a growing family of hybrid Mamba-Transformer architectures (Jamba, Zamba, Hymba, Samba, Nemotron-H) interleaves SSM and attention blocks to retain long-context accuracy at a fraction of the cost. This paper (i) reviews the design space of hybrid Mamba-Transformer architectures; (ii) surveys EEG decoding architectures — EEGNet, DeepConvNet, ATCNet, EEG Conformer, EEG-Deformer, EEGMamba, and Brain2Qwerty — for classifying characters from brain signals; and (iii) describes a Conv + hybrid Mamba-Transformer pipeline for M/EEG-to-text decoding in which a convolutional module feeds a Nemotron-H-style stack of Mamba-2 blocks with periodic global attention, a connectionist temporal classification (CTC) head produces character sequences, a space-aware segmenter pools frames into pseudo-word embeddings aligned to language-model space through a contrastive loss, and a LoRA-adapted 1.1B-parameter large language model (LLM) corrects the decoded text end to end. We compare published character-error rates across modalities and models, analyze why the hybrid inductive bias matches the hierarchical, sustained, and superposed neural representations measured during language production, and specify an evaluation protocol on a 35-participant Spanish M/EEG typing dataset.

Index Terms—Brain-computer interfaces, electroencephalography (EEG), magnetoencephalography (MEG), state-space models, Mamba-2, hybrid Transformer, connectionist temporal classification, large language model, brain-to-text decoding.

I. Introduction

Restoring communication for patients who have lost the ability to speak or move is one of the most impactful goals of neurotechnology. Invasive speech and typing neuroprostheses have reached remarkable performance — for example 90 characters per minute with offline character error rates (CER) below 1% from intracortical handwriting signals^[21] and word error rates (WER) of 9.1% (50-word vocabulary) to 23.8% (125,000-word vocabulary) from intracortical speech signals^[22] — but they require neurosurgery. Non-invasive decoding is far safer yet historically much weaker: EEG letter decoding over a 10-letter alphabet was reported at a CER of 75.8%^[29], and perception-side decoding of speech from MEG reached 41% top-10 accuracy over a limited vocabulary^[20]. Brain2Qwerty recently narrowed this gap by decoding the production of briefly memorized, typed sentences from M/EEG, reaching a CER of 32% with MEG and 67% with EEG using a three-stage architecture: a convolutional module on 500 ms windows, a sentence-level Transformer, and a pretrained 9-gram character language model (LM) that corrects the decoded text^[18].

Two bottlenecks remain in this line of work. First, the sentence-level Transformer scales quadratically with sequence length, which limits context, batch size, and on-device deployment for long sentences. Second, a shallow character n-gram LM can only correct local orthographic errors; it cannot exploit sentence-level semantics, so word-level error correction is weak precisely where non-invasive signals are noisiest.

Meanwhile, sequence modeling has been reshaped by selective state-space models. Mamba^[3] and Mamba-2^[4] match or approach Transformer quality with linear-time recurrent inference, and hybrid architectures that interleave a small number of attention layers into an SSM backbone — MambaFormer^[9], Jamba^[5], Zamba^[6], Samba^[8], Hymba^[7], and the production-scale Nemotron-H family^[10] — recover the few-shot recall abilities of attention while keeping SSM efficiency. In the EEG domain specifically, EEGMamba demonstrated that bidirectional Mamba backbones outperform attention-based classifiers on long recordings with linear memory growth^[16], and a foundation-model variant pretrained Mamba encoders on large EEG corpora^[17].

This paper connects these two threads. We argue that the natural architecture for brain-to-text decoding is a **Conv + hybrid Mamba-Transformer**: convolutions extract local spatio-temporal features, Mamba-2 blocks propagate information across long sentences in linear time, sparse global attention blocks recover exact long-range alignment, and an LLM — rather than an n-gram — performs word-level correction. The argument is also neuroscientifically motivated: during typed language production, letter-, syllable-, word-, and context-level representations are sustained far beyond their execution time, superposed in population activity, and refreshed at level-dependent speeds^[19]. A decoder must therefore integrate local evidence over windows of very different durations, which is precisely the regime where hybrid SSM-attention stacks excel.

Contributions. (i) A structured review of hybrid Mamba-Transformer architectures and their fusion patterns (Section II). (ii) A comparison of EEG character-classification architectures — EEGNet, DeepConvNet, ATCNet, EEG Conformer, EEG-Deformer, EEGMamba, and Brain2Qwerty — emphasizing the trade-off between local feature extraction and global sequence modeling (Section III). (iii) The description of a Conv + hybrid Mamba-2/attention encoder with a CTC character head, word-level contrastive alignment, and end-to-end LoRA-adapted LLM correction, implemented as Brain2Qwerty V3 (Section IV). (iv) An analysis of language-model correction strategies from n-grams to LLMs (Section V) and a literature-anchored comparative evaluation protocol on a Spanish M/EEG typing dataset (Section VI).

II. Background: State-Space Models and Hybrid Architectures

A. Structured State-Space Models and Mamba

Continuous-time linear state-space models (SSMs) map an input signal $x(t)$ to an output $y(t)$ through a latent state $h(t)$:

$$\dot{h}(t) = A h(t) + B x(t), \quad y(t) = C h(t) \quad (1)$$

After zero-order-hold discretization with step size Δ , the system becomes a linear recurrence that can be computed either sequentially in $O(T)$ time or as a parallel scan during training:

$$\tilde{A} = \exp(\Delta A), \quad \tilde{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \Delta B, \quad h_t = \tilde{A} h_{t-1} + \tilde{B} x_t \quad (2)$$

Structured SSMs such as S4^[2] make this recurrence trainable at scale by imposing structure on A . Mamba^[3] adds a *selection mechanism*: the matrices B , C and the step size Δ become functions of the current input,

$$B_t = W_B x_t, \quad C_t = W_C x_t, \quad \Delta_t = \text{softplus}(W_\Delta x_t + b_\Delta) \quad (3)$$

so the model decides per time step what to remember or forget — a content-aware gating that is particularly attractive for EEG, where informative evoked responses are sparsely embedded in background noise. Because the recurrence is time-varying, the convolutional view of S4 no longer applies; Mamba therefore uses a hardware-aware parallel scan.

B. Mamba-2 and the State-Space Duality

Mamba-2 (structured state-space duality, SSD)^[4] shows that selective SSMs and masked linear attention are two parametrizations of the same operation. The recurrence $h_t = \exp(\Delta_t A) h_{t-1} + \Delta_t B_t x_t$ admits a quadratic dual form $y = (L \cdot C B^T) X$ (element-wise product in the mask L), where the decay mask $L[t, s] = \exp(\sum_{u=s+1..t} \Delta_u A)$ plays the role of a causal attention mask with exponential forgetting. This duality is what makes hybrid designs principled rather than ad hoc: a Mamba-2 block is attention with a fixed, input-modulated exponential kernel, while a softmax attention block is a data-dependent kernel without forgetting. Stacking both covers complementary inductive biases. SSD further splits the channel dimension into heads and shares the state across channels within a head, which raises arithmetic intensity and allows block-decomposed (chunked) computation that is 2-8x faster than the original selective scan while remaining exactly recurrent at inference time^[4].

C. Hybrid Mamba-Transformer Architectures

Pure SSMs are efficient but weaker than attention on tasks that require exact recall of rare token pairs, and pure attention is expensive at long contexts. Hybrid architectures resolve this tension in several ways (Table I). **MambaFormer**^[9] interleaves Mamba and attention blocks in a single stack and shows improved in-context learning. **Jamba**^[5] interleaves Transformer and Mamba layers at a 1:7 ratio, adds mixture-of-experts MLPs, and reaches 256K-token contexts with an 8x smaller key-value (KV) cache than a vanilla Transformer. **Zamba**^[6] attaches a single shared global attention module every few blocks to a Mamba backbone, amortizing attention parameters. **Samba**^[8] combines Mamba with sliding-window attention (M-MLP-SWA-MLP repeats) and extrapolates to 1M-token contexts. **Hymba**^[7] fuses attention heads and SSM heads *in parallel within the same layer* (hybrid-head), adding learnable meta tokens that act as a compressed cache. At production scale, **Nemotron-H**^[10] uses a Mamba-2-majority stack with only ~8% attention blocks and reports accuracy comparable to similarly sized Transformers with higher throughput. Two findings from this literature matter here: (i) very few attention blocks suffice when the SSM carries the bulk of sequence propagation, and (ii) positional embeddings can be dropped in Mamba blocks because the recurrence is order-aware^{[5][10]}.

TABLE I
Hybrid Mamba-Transformer Architectures (Language Modeling)

Model	Year	Fusion pattern	Attention role
MambaFormer ^[9]	2024	Interleaved Mamba / attention blocks	In-context recall
Jamba ^[5]	2024	Interleaved 1:7 attn:Mamba + MoE MLPs	1/8 of layers; 256K ctx, 8x smaller KV cache
Zamba ^[6]	2024	Mamba backbone + one shared attention	Single shared global module
Samba ^[8]	2024	Mamba + sliding-window attention repeats	Local (windowed) recall; 1M ctx
Hymba ^[7]	2024	Parallel attention + SSM heads per layer	High-resolution recall heads
Nemotron-H ^[10]	2025	Mamba-2 majority, ~8% attention blocks	Sparse global recall; production scale

III. EEG Architectures for Character-Level Decoding

A. Convolutional Baselines

CNNs remain the default inductive bias for EEG because electrode arrays are spatially structured and evoked responses are temporally localized. **EEGNet**^[11] packages this bias compactly: a temporal convolution, a depthwise spatial convolution across channels, and separable convolutions, in a few thousand parameters. **DeepConvNet** and **ShallowConvNet**^[12] generalize filter-bank decompositions with deeper convolutional stacks. These models excel at short segments but have a bounded receptive field and no mechanism for sentence-level context, which limits them in typed-text decoding: in Brain2Qwerty's comparison, EEGNet was 1.14x worse (EEG) and 2.25x worse (MEG) in CER than the full Conv+Transformer+LM model^[18].

B. Convolution-Transformer Hybrids

EEG Conformer^[13] is the canonical conv-transformer for EEG: a convolutional module extracts local temporal and spatial features, and a self-attention encoder captures global correlations, reaching 78.66% (BCI Competition IV-2a, hold-out), 84.63% (IV-2b) and 95.30% (SEED) in its original evaluation. **ATCNet**^[15] augments a temporal convolutional network with multi-head attention for motor imagery. **EEG-Deformer**^[14] adds dense coarse-to-fine temporal branches to the Conformer pattern, improving over EEG Conformer by 2.9-5.0 percentage points across attention, fatigue, and workload classification. The consistent lesson of this family is that convolutions plus attention beat either alone — but all of them inherit the $O(T^2)$ cost and KV-cache growth of softmax attention, and all are trained as isolated classifiers rather than as the front-end of a text production pipeline.

C. State-Space EEG Models

EEGMamba^[16] was the first architecture to bring selective SSMs to EEG classification: a spatio-temporal-adaptive tokenizer, bidirectional Mamba blocks (necessary because offline EEG analysis is non-causal), and a task-aware mixture of experts. Across eight public datasets spanning seizure detection, emotion recognition, sleep staging, and motor imagery it outperformed EEGNet, AttnSleep, and EEG Conformer on most benchmarks, while memory grows linearly with signal length — handling sequences beyond 40,000 samples where attention baselines approach out-of-memory^[16]. A successor scaled the idea into an EEG foundation model with patch-based masked reconstruction pretraining^[17]. These results establish that SSMs are competitive with attention on EEG; what they do not address is open-vocabulary text production, which additionally requires character-level alignment (CTC-style) and linguistic correction.

D. Brain2Qwerty: Brain-to-Text via Typing

Brain2Qwerty^[18] is the reference point for this paper. Thirty-five healthy participants read Spanish sentences presented word by word (rapid serial visual presentation), memorized them during a 1.5 s fixation, and typed them from memory without visual feedback, while 306-channel MEG and/or 64-channel EEG were recorded. The decoder combines (1) a convolutional module with a subject-specific linear layer applied to 500 ms windows, (2) a sentence-level Transformer over the pooled window embeddings, and (3) a pretrained 9-gram character LM applied at decoding. Trained end-to-end (about 400M parameters: 258M convolutional + 138M Transformer), it reaches CER = $32 \pm 0.6\%$ (MEG) and $67 \pm 1.5\%$ (EEG), with the best participant at 19% CER on MEG and several perfectly decoded held-out sentences^[18]. Ablations show each stage contributes: the conv module alone already beats EEGNet, the Transformer improves CER significantly, and the LM improves it further. The companion neuroscience analysis^[19] shows that during typing, context-, word-, syllable-, and letter-level representations rise in top-down order and are each sustained much longer than their execution time (letters: about 103 ms keypresses decodable for over 1 s; context sustained for seconds), with up to five successive letters simultaneously decodable — direct evidence that a brain-to-text decoder faces a long, superposed, multi-scale temporal integration problem.

TABLE II
EEG Architectures Relevant to Character-Level Decoding

Model	Year	Sequence core	Global model.	Scaling
EEGNet ^[11]	2018	Depthwise/separable CNN	None	O(T)
DeepConvNet ^[12]	2017	Deep CNN	None	O(T)
ATCNet ^[15]	2023	Attention + TCN	Attention over TCN feats.	O(T^2)
EEG Conformer ^[13]	2023	Conv module + Transformer enc.	Full self-attention	O(T^2)
EEG-Deformer ^[14]	2024	Dense coarse-to-fine conv-Transformer	Full self-attention	O(T^2)
EEGMamba ^[16]	2024	Bi-directional Mamba + MoE	Selective SSM (bi-dir.)	O(T)
Brain2Qwerty ^[8]	2025	Conv + sentence Transformer + 9-gram LM	Full self-attention	O(T^2)
Ours (B2Q V3)	2026	Conv + hybrid Mamba-2/attention + LoRA LLM	SSM + sparse attention	mostly O(T)

IV. Proposed Conv + Hybrid Mamba-Transformer Pipeline

A. Overview

The proposed architecture (Fig. 1), implemented as Brain2Qwerty V3, replaces the sentence-level Transformer of Brain2Qwerty with a Nemotron-H-style hybrid Mamba-2/attention stack^[10] and replaces the shallow 9-gram corrector with an end-to-end LoRA-adapted LLM^[27]. Five modules share one encoder forward pass: (i) a convolutional front-end; (ii) the hybrid sequence core; (iii) a CTC character head with auxiliary supervision; (iv) a CTC space-based word segmenter whose pooled word embeddings are aligned to the LLM's embedding space by a contrastive loss; and (v) a frozen 1.1B-parameter LLM^[32] with LoRA adapters (0.05% trainable parameters) that generates the corrected sentence conditioned on both the greedy CTC transcript and the neural word embeddings. The total objective is

$$\mathcal{L} = w_{\text{ctc}} \mathcal{L}_{\text{CTC}} + w_{\text{con}} \mathcal{L}_{\text{con}} + w_{\text{llm}} \mathcal{L}_{\text{LM}} \quad (4)$$

with weights (1 - alpha - beta, alpha, beta), alpha = 0.1, beta = 0.01 by default, gated by epoch so that each loss can be phased in after the encoder has stabilized.

B. Convolutional Front-End

Following the Brain2Qwerty design^[18] and the subject layer of Defossez et al.^[20], each 500 ms M/EEG window (band-passed 0.5-45 Hz, notch-filtered at line-noise harmonics, downsampled) is linearly remapped by a subject-specific layer that learns a per-participant spatial montage, followed by spatial and temporal convolutions with downsampling. The output is a sequence of window embeddings at a reduced frame rate, short enough that a handful of attention blocks is cheap, yet long enough that the SSM blocks carry the bulk of temporal propagation. This division of labor — convolutions for local spatio-temporal features, SSM for long-range integration — mirrors the evidence that conv+sequence hybrids dominate pure CNNs and pure Transformers on EEG^{[13], [14]}.

C. Hybrid Mamba-2/Attention Sequence Core

The sequence core stacks $N = 8$ pre-norm residual blocks in the pattern [M, M, M, A, M, M, M, A] (one attention block every four), followed by a final RMSNorm. Each Mamba-2 block applies the SSD mixer: an input projection produces the gate z , the conv branch $[x, B, C]$, and the step-size logits; a causal depthwise convolution (kernel 4) with SiLU pre-smooths the branch — acting as a learned local filter against high-frequency EEG artifacts — and the selective recurrence

$$h_t = \exp(\Delta_t A) h_{t-1} + \Delta_t B_t x_t, \quad y_t = C_t h_t + D x_t \quad (5)$$

is evaluated in its quadratic dual form (Section II-B) computed in float32 and chunked over heads to bound memory. The output passes through a gated RMSNorm (gate applied as $x * \text{silu}(z)$) before the norm, matching the reference implementation) and an output projection. Attention blocks are single x-transformers encoder layers with rotary position embeddings^[33], RMSNorm, four heads, and a small feed-forward sublayer. No positional information is added to Mamba blocks, since the recurrence is order-aware^[5]. With the frame rate produced by the conv front-end (a few hundred frames per sentence), the 75% of blocks that are SSM run in O(T), and only the two attention blocks pay the quadratic cost; Fig. 2(a) illustrates the scaling argument. The encoder totals approximately 358M trainable parameters — comparable to the Brain2Qwerty Conv+Transformer, so any gain can be attributed to architecture rather than scale.

D. CTC Character Decoding

Because keypress timings within the continuous typing stream are only partially aligned with neural responses, the character head is trained with connectionist temporal classification (CTC)^[26], which marginalizes over all alignments π between encoder frames and the target character sequence y :

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in B^{-1}(y)} \prod_t p(\pi_t | X) \quad (6)$$

An auxiliary CTC head on an intermediate representation is blended with the final head (30% final, 70% auxiliary by default), which stabilizes training of the deep hybrid stack. At inference, greedy CTC decoding yields a raw character transcript (e.g. 'stamistosiasigue la distribucion').

E. Word Segmentation and Contrastive Alignment

To move from characters to words, predicted space characters in the CTC posteriors define boundaries; encoder frames between successive spaces (including blank frames) are mean-pooled by a small trainable pooler into one pseudo-word embedding per hypothesized word. A projection adapter maps these embeddings into the LLM's word-embedding space, where a word-level contrastive loss pulls each neural word embedding toward the text embedding of the true word (computed once by the frozen LLM) and away from the other words in the batch. This gives the encoder a word-level training signal that CTC — a purely character-level objective — cannot provide, and it is what makes the neural word embeddings interpretable to the LLM at correction time.

F. LLM-Based Word Correction

The final stage replaces the 9-gram character LM of Brain2Qwerty^[18] with a LoRA-adapted^[27] instruction-tuned LLM (1.1B parameters, frozen base; 563K trainable adapter parameters plus a 2.1M projection adapter). The model receives a prefix built in embedding space: the tokenized greedy CTC transcript ('CTC: ...'), followed by the adapted neural word embeddings ('MEG: ...'), and a response marker ('Output:'), and is trained with teacher-forced cross-entropy (label smoothing 0.02) to generate the true sentence. At inference, beam search (16 beams, length penalty 0.2) produces the corrected sentence. Because the LLM sees both a noisy character transcript and a word-level neural representation, it can arbitrate between them: character errors that would defeat a character n-gram (e.g. 'EK BENEFUCU SYOERA KIS RUESGIS' for 'el beneficio supera los riesgos') are exactly the regime where a pretrained LLM's lexical and syntactic priors are strongest^{[18], [31]}. Modality dropout on both the neural embeddings and the CTC text during training prevents the LLM from ignoring the neural input.

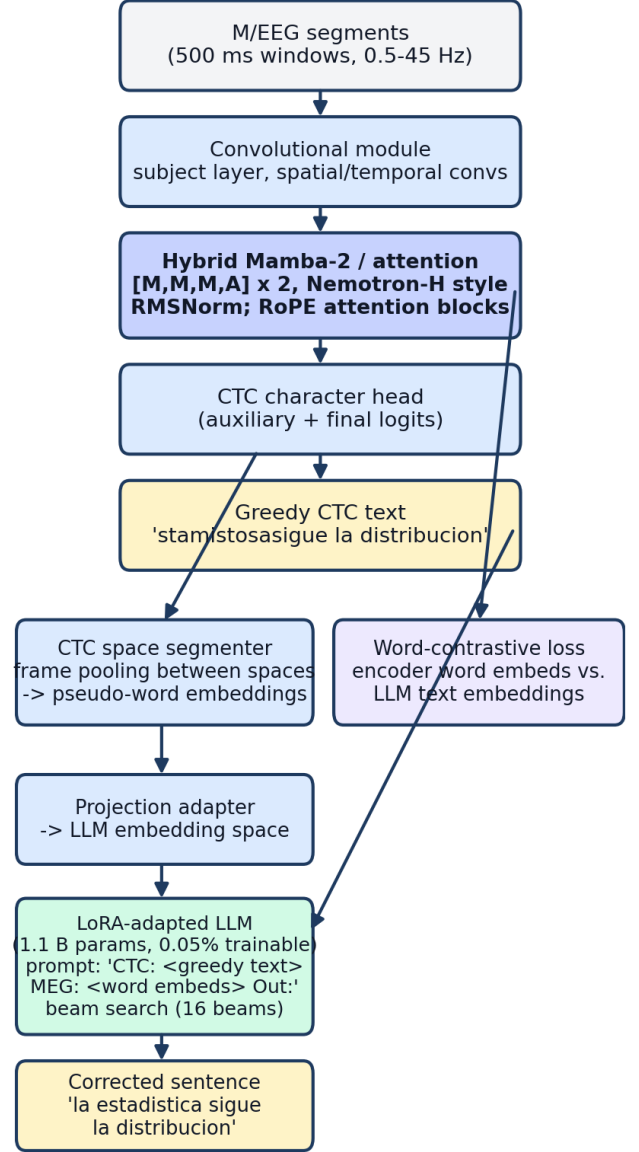


Fig. 1. Proposed Conv + hybrid Mamba-Transformer brain-to-text pipeline (Brain2Qwerty V3). A convolutional front-end feeds a Nemotron-H-style stack of Mamba-2 blocks with periodic attention; a CTC head yields character sequences; a space-based segmenter pools frames into pseudo-word embeddings aligned to LLM space by a contrastive loss; a LoRA-adapted LLM generates the corrected sentence.

V. Language-Model Correction Strategies for Brain-to-Text

Neural decoders and language models play complementary roles: the decoder maps brain activity to symbols, and the LM enforces linguistic plausibility. Four strategies exist, in increasing order of LM capacity (Table III). (1) **Character n-gram fusion**: Brain2Qwerty rescues CTC outputs with a pretrained 9-gram character LM^[18], similar in spirit to the 5-gram decoders used in invasive speech BCIs^[22]. n-grams are cheap and robust, but their short context cannot model semantics. (2) **LLM rescoring**: n-best hypotheses from the decoder are rescored by a pretrained LLM; in speech BCI pipelines, adding conversational context to OPT- or GPT-2-scale rescoring reduced both CER and WER by roughly 2 absolute points relative to n-gram-only decoding^[22]. (3) **Sequence-to-sequence translation**: DeWave maps discrete EEG codex tokens to text with a BART encoder-decoder^[24], building on open-vocabulary EEG-to-text with pretrained LM priors^[25]; this works for reading-evoked EEG but entangles alignment and correction in a single black box. (4) **End-to-end conditioned LLM (ours)**: the LLM is fine-tuned with LoRA to read both the noisy transcript and the neural word embeddings, so correction is trained jointly with the encoder and can exploit information that never survives greedy character decoding — an approach consistent with the broader trend of fusing LLMs into BCI spellers for word prediction, completion, and error correction^[31]. The Word Error Rate (WER) and semantic error metrics (SemER) are the natural evaluation criteria at this stage.

TABLE III
Language-Model Correction Strategies for Brain-to-Text

Strategy	Example	Strength	Weakness
Character n-gram fusion	Brain2Qwerty 9-gram ^[18]	Cheap, robust	No semantics; short context
LLM rescoring of n-best	OPT/GPT-2 rescoring ^[22]	No retraining	Limited by n-best quality
Seq2seq translation	DeWave (BART) ^[24] , ^[25]	Open vocabulary	Entangles alignment and correction
End-to-end LoRA LLM (ours)	B2Q V3 ^[27] , ^[32]	Sees neural word embeds + transcript	Heavier; hallucination risk

VI. Experimental Setup and Comparative Analysis

A. Dataset and Task

Experiments use the Spanish read-wait-type dataset recorded at the Basque Center on Cognition, Brain and Language (BCBL)^[18], ^[19]: 35 healthy native Spanish speakers (20 MEG, 20 EEG, 5 both), 128 unique declarative sentences of 5-8 words, MEG at 306 channels (1 kHz, online 0.1-330 Hz) and EEG at 61+3 channels, preprocessed to 0.5-45 Hz and downsampled. Each trial consists of RSVP reading, a 1.5 s wait, and memory-guided typing with no visual letter feedback. Typographical errors are filtered by Levenshtein alignment between the displayed and typed sentences.

B. Metrics

Character error rate $CER = (S + D + I)/N$ (substitutions, deletions, insertions over sentence length) is the primary metric, following the ASR convention adopted in brain-to-text work^[18], ^[21]; word error rate (WER) and SemER evaluate the corrected output. Statistical comparisons across subjects use Wilcoxon signed-rank tests, as in the reference study.

C. Literature-Reported Comparison

Fig. 2(b) and Table IV collect the published reference points. Non-invasive letter decoding without a sentence context is weak: Crell and Muller-Putz report 75.8% CER over a 10-letter EEG alphabet^[29]. Sentence-level decoding with a deep pipeline changes the picture: Brain2Qwerty reaches 67% CER on EEG and 32% on MEG, and 19% for the best MEG participant — at which point some held-out sentences are decoded perfectly^[18]. For calibration, invasive systems remain well ahead: 9.1% WER (50-word) and 23.8% WER (125,000-word) for intracortical speech^[22], 15.2% CER at 79 words per minute for ECoG speech^[23], and under 1% offline CER for intracortical handwriting with a correction model^[21]. Two implications follow: the sensor modality contributes about a 2x CER gap (MEG vs. EEG), and architecture plus language modeling contributes a comparable factor within a modality — the Conv+Transformer+LM pipeline improves CER 1.14x (EEG) and 2.25x (MEG) over EEGNet, and ablations show the Transformer and the LM each add significant gains^[18].

D. Expected Benefits of the Hybrid Design and Evaluation Protocol

The proposed V3 pipeline is currently being trained on the Spanish BCBL dataset; consequently, we do not report new empirical numbers and instead state the hypotheses the design targets. **H1 (efficiency)**: at equal parameters (~358M), replacing 75% of attention blocks with Mamba-2 reduces memory and increases throughput for sentence-length sequences, following the scaling behavior measured for SSM EEG backbones^[16] and hybrid LLMs^[5], ^[10]. **H2 (accuracy)**: the selective SSM's input-dependent gating should track the sustained, superposed letter/word representations documented in the companion neuroscience study^[19] better than full attention at equal compute, because exponential decay with selective gating matches the measured dynamic-code structure. **H3 (correction)**: end-to-end LoRA-adapted LLM correction should reduce WER beyond the 9-gram LM of the reference system, particularly on rare and out-of-vocabulary words, where n-gram priors are weakest^[18], ^[31]. The evaluation protocol mirrors Brain2Qwerty: per-subject and pooled models, held-out sentences, Wilcoxon tests against the V2 Conv+Transformer+9-gram baseline, ablations removing attention blocks, the contrastive loss, and the LLM stage, and complexity benchmarks (memory and latency vs. sentence length).

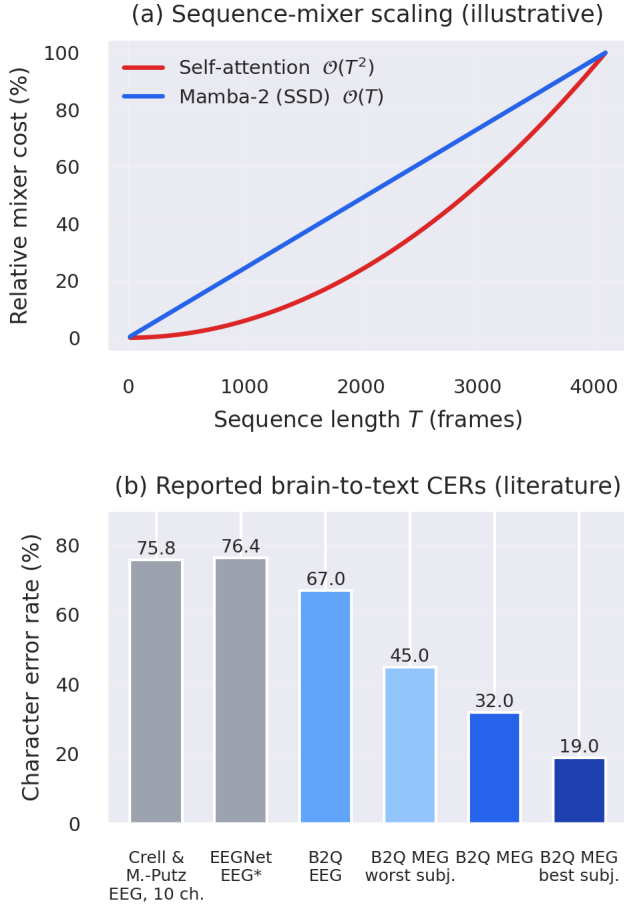


Fig. 2. (a) Illustrative scaling of the sequence mixer: softmax attention grows quadratically in sequence length, while the Mamba-2 SSD recurrence grows linearly^{[3], [4]}. (b) Published brain-to-text character error rates^{[18], [29]}. B2Q = Brain2Qwerty. *As reported in^[18]. †Derived from the reported 1.14x CER improvement of the full Brain2Qwerty model over EEGNet on EEG.

TABLE IV
Reported Brain-to-Text / Typing Decoding Performance

System	Modality	Task	Reported error
Crell and Muller-Putz ^[29]	EEG	10-letter classification	75.8% CER*
EEGNet baseline ^{[11], [18]}	EEG	Sentence typing	~76% CER†
Brain2Qwerty ^[18]	EEG	Sentence typing	67 +/- 1.5% CER
Brain2Qwerty ^[18]	MEG	Sentence typing	32 +/- 0.6% CER (best 19%)
Willet 2021 ^[21]	Intracortical	Handwriting	<1% CER (offline, w/ LM)
Willet 2023 ^[22]	Intracortical	Speech	9.1% WER (50-word); 23.8% (125k)
Metzger 2023 ^[23]	ECoG	Speech	15.2% CER, 79 wpm

VII. Discussion

Why hybrids fit brain-to-text. The measured neural code of typing is hierarchical and superposed: fast, dynamic letter codes co-exist with slow, sustained context codes^[19]. Convolutional layers match the fast motor-somatosensory transients; Mamba-2's input-dependent decay implements multi-scale memory in a single block type, with different heads learning different time constants through the step-size parameter; and sparse attention blocks recover exact alignment when a distant frame must be compared directly — the same reason hybrids dominate pure SSMs on recall-heavy language tasks^{[7], [9]}.

Practicality. The hybrid stack keeps the parameter count of the Transformer baseline while moving most sequence computation into linear-time blocks, and the LoRA correction stage adds only ~0.05% trainable parameters over a frozen 1.1B model — the entire system trains in bfloat16 on commodity GPUs. Because only the correction stage requires the LLM at inference, latency-critical online decoding can still use greedy CTC output while the LLM refines text at sentence boundaries.

Limitations. (i) Non-invasive decoding remains motor-dominated: error analyses in the reference study show decoding relies substantially on motor processes, so applicability to non-typing patients (the clinical target) is unproven^[18]. (ii) The dataset is small (35 participants, 128 sentences), and hybrid LLM findings may not transfer at this scale. (iii) Our implementation evaluates the SSD dual form in its quadratic mode; the memory benefit of the chunked scan is realized only at longer contexts. (iv) LLM correction can hallucinate fluent but wrong sentences, so CER/WER must be reported alongside semantic metrics, and high-stakes use needs conservative operating points^[31]. (v) Causality: for real-time use the bidirectional/non-causal components must be restricted, echoing the distinction EEGMamba draws between offline and online settings^[16].

Future work. Promising directions include bidirectional Mamba-2 variants for offline decoding^[16], pretraining the encoder on large unlabeled EEG corpora in the style of EEG foundation models^[17], scaling data collection (the reference study found CER improves log-linearly with recording time^[18]), cross-subject transfer through the subject layer, and unifying the LLM correction with conversational context as in rescaling-based speech BCI pipelines^[22].

VIII. Conclusion

Hybrid Mamba-Transformer architectures reconcile the two dominant sequence-model paradigms: SSMs give linear-time, order-aware propagation with multi-scale forgetting, while sparse attention restores exact global recall. For EEG-to-text decoding, this combination maps naturally onto the hierarchical, sustained, and superposed neural representations of language production. The presented Conv + hybrid Mamba-2/attention pipeline — convolutional front-end, Nemotron-H-style core, CTC character head, contrastively aligned word embeddings, and LoRA-adapted LLM correction — upgrades every stage of the Brain2Qwerty reference architecture at comparable parameter count, and defines a concrete, testable path toward accurate non-invasive brain-to-text systems. The literature comparison underscores the stakes: closing even part of the gap between the 32% MEG CER of current non-invasive decoding and the single-digit error rates of invasive neuroprostheses would mark a qualitative step toward safe communication BCIs.

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